

Vinay Reddy Oppuri

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Summary

Full Stack Engineer with hands-on experience building AI-integrated web applications and LLM-powered automation pipelines. Proficient in TypeScript, React, Next.js, and Node.js, with practical exposure to Generative AI workflows including RAG pipelines, LLM APIs, and vector-based semantic search. Comfortable working across the stack, from type-safe backend services and cloud deployments to responsive, performant frontends.

Technical Skills

Languages: JavaScript, TypeScript, Python, C, C++

Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, shadcn/ui

Backend: Node.js, tRPC, REST APIs, FastAPI, Uvicorn, Pydantic

AI & GenAI: LLM APIs, RAG Pipelines, Hugging Face, Vector Embeddings

Databases: PostgreSQL (NeonDB), MongoDB, Supabase

Tools & DevOps: Git, GitHub, Vercel, Render, Docker, Turborepo, Drizzle ORM

Experience

Website Development Team Member

IIT Raichur

Official Institute Websites — iiitr.ac.in

- Maintain and enhance multiple official web platforms under the `iiitr.ac.in` domain, contributing to their modernization with a scalable **Next.js App Router** architecture.
- Developed the **Training & Placement Cell Portal** (`tnp-iiitr.vercel.app`) using **Next.js** and **Tailwind CSS**.
- Architected reusable and scalable components, improving development efficiency and maintainability across projects.

Projects

Rescomail [Next.js, TypeScript, Turborepo, Python, FastAPI, Gemini API, Trigger.dev v3, Tavily, Drizzle ORM, Better Auth, PostgreSQL, Vercel, Render]

- Architected a full-stack AI job copilot in a Turborepo monorepo, pairing Next.js with a FastAPI microservice deployed on Vercel and Render with shared-secret authentication.
- Built an ATS intelligence pipeline using Gemini embeddings and cosine similarity, implementing a RAG workflow for skill-gap analysis and compatibility scoring.
- Developed cold email generation via Gemini with automated model failover, Pydantic schemas, and Tavily web scraping for dynamic company context enrichment.
- Designed a type-safe data layer with Drizzle ORM on PostgreSQL, integrated with Better Auth, UploadThing, and a Trigger.dev (v3) background task queue.

Mono

[Next.js, TypeScript, Node.js, tRPC, PostgreSQL, Drizzle ORM, Better Auth, Gemini AI]

- Built an AI agent workspace where users create task-specific LLM-powered assistants to automate repetitive workflows, targeting AI-driven business automation use cases.
- Integrated the Gemini LLM API to deliver context-aware responses grounded in saved conversation history, implementing a lightweight RAG-style retrieval pattern for recurring task automation.
- Engineered type-safe backend APIs using Node.js, tRPC, and Drizzle ORM on PostgreSQL with user-scoped access control and Better Auth session management.

Education

in Artificial Intelligence & Data Science

Indian Institute of Information Technology, Raichur

JEE Mains 2023: Ranked in top 4.5% nationally

B.Tech

Expected 2027